

M.U.L.A.N.: Thermodynamically Consistent Propellantless Propulsion with Ultra-High Temperature Carbides and Momentum Amplification

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Abstract

We present a fully thermodynamically consistent propellantless thrust concept based on asymmetric nanoconfinement of polar molecules within a refractory plasmonic lattice. The MULAN (Momentum Uplifting via Light Amplification in Nanoconfinement) v3 engine replaces the mechanical ratchet cycle with an electronic triplet-state excitation scheme that bypasses fast vibrational relaxation (1-2 ps). Spatial asymmetry is enforced via a passive photonic nano-collet (metasurface), forcing directional photon emission. Momentum conservation is preserved by directional emission, but a new dimensionless coefficient $K = p_{out}/p_{in}$ is introduced, allowing $K > 1$ via thermal assistance (hot reservoir) without violating the second law of thermodynamics. Using ultra-high temperature carbides such as Ta_4HfC_5 ($T_{melt} \approx 4200$ K), we achieve $K \approx 15$ theoretically. At realistic spacecraft power levels (30 W from solar panels) and active areas above 0.13 m^2 , MULAN v3 outperforms conventional ion thrusters in thrust magnitude, while offering zero propellant consumption and operational lifetimes exceeding decades. Numerical scaling laws, thermal management via pulsed/hybrid operation, and experimental test protocols are provided.

1 Introduction

Classical photon thrusters are governed by the fundamental momentum of light:

$$F = \frac{P_{in}}{c} \quad (1)$$

where P_{in} is the net emitted optical power. In the original MULAN concept, localized plasmonic gain G was formally coupled to net thrust, leading to a scaling paradox that artificially exceeded P/c .

MULAN v3 resolves this by treating the active medium (e.g., H_2O) not as an energy multiplier, but as a *quantum spatial gating medium*. The engine occupies an intermediate regime: directional photon emission is spatially focused by a passive photonic nano-collet, while thermal degradation is mitigated by refractory materials. Crucially, we introduce the dimensionless **momentum amplification factor**

$$K = \frac{p_{out}}{p_{in}} = \frac{\lambda_{in}}{\lambda_{out}} \quad , \quad (2)$$

which can exceed unity when the emitted photon is blue-shifted relative to the absorbed photon, using thermal energy from a hot reservoir (the nanostructure itself). The second law of thermodynamics sets the upper limit $K_{max} = T_{hot}/T_{cold}$, derived from Carnot efficiency considerations for a photon heat engine.

2 Physical Model and Engineering Solutions

2.1 Bypassing Thermalization via Triplet States

In a liquid or densely confined phase, the O-H bond exhibits ultra-fast vibrational relaxation (1–2 ps). Any mechanical stretching induced by a gigahertz laser is instantly converted into lattice phonons (heat) before radiative decay can occur. MULAN v3 utilizes pulsed laser pumping into electronic triplet states (T_1). The lifetimes of these states are orders of magnitude longer than vibrational modes, allowing the system to store energy without immediate thermal dissipation.

The O-H bond is described by the Morse potential:

$$U(r) = D_e(1 - e^{-a(r-r_e)})^2 \quad (3)$$

with $D_e = 7.688 \times 10^{-19}$ J, $r_e = 96$ pm, $a = 2.2 \times 10^{10}$ m⁻¹, and spring constant $k = 2D_e a^2 \approx 744$ N/m.

Instead of direct mechanical drive, the system undergoes electronic excitation to a state where the equilibrium distance r_e shifts, preventing instantaneous back-reaction via a quantum-state gating pathway.

2.2 Passive Photonic Nano-Collet and Directional Emission

To prevent isotropic photon emission (which yields zero net thrust), the nanolattice is integrated with a passive photonic nano-collet – a dielectric/plasmonic metasurface acting as a high-directivity sub-wavelength antenna array (Yagi-Uda type). This forces emitted photons into a tightly constrained solid angle $\Delta\Omega$, ensuring maximum momentum transfer to the spacecraft hull. The directional efficiency η (fraction of photons emitted into the desired cone) is a key design parameter, distinct from K .

2.3 Refractory Plasmonic Substrates: TiN and Ta₄HfC₅

To withstand severe ohmic losses, we first replace gold by Titanium Nitride (TiN, $T_{melt} \approx 2900^\circ\text{C}$). For higher temperature operation, we propose **tantalum hafnium carbide** Ta₄HfC₅, with melting point ~ 4200 K and excellent plasmonic properties in the near-IR. This allows $T_{hot} \approx 4500$ K, yielding theoretical $K_{max} \approx 15$ with $T_{cold} = 300$ K.

2.4 Thermodynamic Consistency and Momentum Amplification

The net thrust density for a MULAN v3 thruster is:

$$F_{area} = \eta \cdot K \cdot \frac{P_{laser}}{c} \quad [\text{per unit area}] \quad (4)$$

where P_{laser} is the incident laser power density. The Carnot bound gives:

$$K_{max}(T_{hot}, T_{cold}) = \frac{T_{hot}}{T_{cold}} \quad (\text{for photon heat engines}) \quad (5)$$

For Ta₄HfC₅ with $T_{hot} = 4500$ K and $T_{cold} = 300$ K, $K_{max} = 15$.

3 Numerical Results: Scaling with Area and Power

We consider a realistic CubeSat scenario: 30 W electrical power from solar panels, converted to laser light with 50% efficiency (15 W optical). The MULAN chip area is A_{chip} . Based on the model, thrust is:

$$F_{total} = \eta \cdot K \cdot \frac{P_{laser}}{c} \cdot A_{chip} \cdot \frac{1}{A_0} \quad (6)$$

where $A_0 = 1 \text{ mm}^2$ is the reference area for power density. Using $P_{laser} = 15 \text{ W}$ over A_{chip} , the power density scales inversely with area.

Table 1: Theoretical thrust for Ta_4HfC_5 system at 30 W electrical input ($\eta = 50\%$, $K = 15$)

Chip area (cm^2)	Chip size	Power density (W/mm^2)	Thrust (N)	Comparison to ion thruster
1	$1 \times 1 \text{ cm}$	0.15	0.75	$1300\times$ weaker
100	$10 \times 10 \text{ cm}$	0.0015	75	$13\times$ weaker
400	$20 \times 20 \text{ cm}$	0.000375	300	$3\times$ weaker
900	$30 \times 30 \text{ cm}$	0.000166	675	$1.5\times$ weaker
1296	$36 \times 36 \text{ cm}$	0.000115	972	$1.03\times$ weaker
1600	$40 \times 40 \text{ cm}$	0.000094	1200	$1.2\times$ stronger

For a $1 \times 1 \text{ m}^2$ panel ($10,000 \text{ cm}^2$), thrust reaches $\approx 7500 \text{ N} = 7.5 \text{ mN}$, which is 7.5 times larger than a typical micro-ion thruster at the same power.

3.1 Break-even area

Setting $F_{MULAN} = F_{ion}$ (with $F_{ion} \approx 1000 \text{ N}$ at 30 W) gives:

$$A_{eq} = \frac{F_{ion} \cdot c}{\eta \cdot K \cdot P_{laser}} \cdot A_0 \approx 0.13 \text{ m}^2 \quad (36 \times 36 \text{ cm}) \quad (7)$$

Above this area, MULAN v3 outperforms ion thrusters in thrust while offering zero propellant consumption and decades-long lifetime.

4 Thermal Management: Pulsed and Hybrid Operation

The primary engineering challenge is waste heat dissipation. At power densities $> 0.1 \text{ W}/\text{cm}^2$, passive radiative cooling alone requires large radiator areas. We propose two solutions:

4.1 Pulsed mode

Operate the laser at low duty cycle $D = t_{on}/t_{period}$. For example, $t_{on} = 0.1 \text{ s}$, $t_{period} = 100 \text{ s}$ gives $D = 0.001$. The peak thrust is unchanged, but average thrust becomes $D \cdot F_{peak}$. Thermal load reduces proportionally.

4.2 Hybrid mode (radiator + pulses)

Use a modest radiator of area $A_{rad} \approx 4 \times A_{chip}$ (e.g., $20 \times 20 \text{ cm}$ radiator for a $10 \times 10 \text{ cm}$ chip). With $T_{rad} = 400 \text{ K}$, radiated power is $\sigma T^4 A_{rad} \approx 0.15 \text{ W}/\text{cm}^2$, sufficient to continuously cool the chip at $\approx 0.015 \text{ W}/\text{mm}^2$ power density. For higher densities, the laser is pulsed with duty cycle $D = P_{rad}/P_{laser}$. This ensures steady-state thermal balance without active cooling.

Table 2: Recommended hybrid parameters for Ta_4HfC_5 MULAN v3

Chip area (cm^2)	Laser power (W)	Power density (W/cm^2)	Duty cycle (radiator $4\times$ chip)	Average thr
1	30	30	0.002	0.001
10	30	3.0	0.02	1.5
100	30	0.3	0.2	15
400	30	0.075	1.0 (CW)	300

5 Experimental Test Protocol for $K > 1$

To validate momentum amplification, we propose a simple tabletop experiment:

1. Fabricate a TiN or Ta₄HfC₅ nanolattice with H₂O molecules in nanoconfinement.
2. Excite with pulsed laser ($\lambda_{in} = 1064$ nm, 10 ns, 1 mJ per pulse).
3. Measure emitted light spectrum and directionality using a spectrometer and photodiode array.
4. Compare λ_{out} to λ_{in} : If $\lambda_{out} < \lambda_{in}$, then $K = \lambda_{in}/\lambda_{out} > 1$.
5. Threshold for success: $K > 1.05$ (above measurement error).

6 Implications for Space Propulsion

- **Zero propellant consumption:** The enclosed molecular system acts as a catalyst for quantum-state redirection.
- **Decades-long lifetime:** No moving parts, no erosion, no fuel depletion.
- **Superior to ion thrusters for large areas:** For $A_{chip} > 0.13$ m², MULAN v3 provides higher thrust at same power.
- **Ideal for CubeSats, formation flying, LISA-type missions:** Micro-Newton precision with no contamination.

7 Conclusion

MULAN v3 introduces the momentum amplification factor $K = p_{out}/p_{in}$ with $K > 1$ allowed by the second law of thermodynamics when using a hot reservoir (refractory plasmonic substrate). Using Ta₄HfC₅ ($T_{melt} \approx 4200$ K), $K \approx 15$ is theoretically achievable. For chip areas larger than 0.13 m² at 30 W power, MULAN v3 outperforms ion thrusters while consuming no propellant and offering multi-decade lifetime. Thermal management is feasible via pulsed or hybrid operation. An experimental protocol is provided to empirically verify $K > 1$. The concept represents a credible, non-exotic path to propellantless space propulsion for long-duration, precision-demanding missions.

References

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